

We define super digit of an integer x using the following rules:

Given an integer, we need to find the *super digit* of the integer.

- If x has only 1 digit, then its super digit is x .
- Otherwise, the super digit of x is equal to the super digit of the sum of the digits of x .

For example, the super digit of **9875** will be calculated as:

super_digit(9875)	9+8+7+5 = 29
super_digit(29)	2 + 9 = 11
super_digit(11)	1 + 1 = 2
super_digit(2)	= 2

Example

$n = '9875'$
 $k = 4$

The number p is created by concatenating the string n k times so the initial $p = 9875987598759875$.

```

superDigit(p) = superDigit(9875987598759875)
                9+8+7+5+9+8+7+5+9+8+7+5+9+8+7+5 = 116
superDigit(p) = superDigit(116)
                1+1+6 = 8
superDigit(p) = superDigit(8)

```

All of the digits of p sum to **116**. The digits of **116** sum to **8**. **8** is only one digit, so it is the super digit.

Function Description

Complete the function *superDigit* in the editor below. It must return the calculated super digit as an integer.

superDigit has the following parameter(s):

- *string n*: a string representation of an integer
- *int k*: the times to concatenate n to make p

Returns

- *int*: the super digit of n repeated k times

Input Format

The first line contains two space separated integers, n and k .

Constraints

- $1 \leq n < 10^{100000}$
- $1 \leq k \leq 10^5$